Individual Contribution Journal: Lab 06

Course: ITAI-1371

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ARINZE OKAFOR

Team Coordination & Leadership

- Scheduled and led our group meetings, setting clear agendas for each session.
- Tracked milestones and followed up with team members to keep everyone on schedule.
- Collected, merged, and finalized all code, markdown notes, and deliverables for submission.

Technical Implementation

- Debugged and resolved notebook syntax errors in the initial setup cells.
- Built and trained the LinearRegression pipeline (Age, Pclass → Fare), achieving an MSE of 3364.92 (RMSE ≈ $58.01).
- Encoded categorical features (Sex → binary) and trained the LogisticRegression model, hitting 74.83% accuracy.
- Extracted and evaluated model weights (.coef_) and completed the Knowledge Check analysis.
- Ran final checks across all cell outputs to verify notebook execution.

Hamza Ajmal

Module 06 Lab – Contribution Journal

Regression and Classification

Notebook setup and data preparation

I reviewed the setup cell that loads the Titanic dataset, removes rows with missing Age for this exercise, selects Age and Pclass as regression features, and splits the data into training and testing sets. I checked that the feature columns and target column were selected as intended.

Regression implementation

I completed the LinearRegression task by creating the model, fitting it with the training data, predicting Fare values for the test data, and calculating mean squared error. I also reviewed the coefficients and intercept to understand how the fitted model represents the relationship between the input features and Fare.

Classification implementation

I reviewed the encoding of Sex into numerical values and the use of Age, Pclass, and Sex to predict Survived. I completed the LogisticRegression task, generated test-set predictions, and evaluated them with accuracy_score.

Verification and concepts

I checked that the workflow separates training data from test data and that each evaluation metric matches the task: mean squared error for a continuous target and accuracy for a class label. I also reviewed the knowledge-check answers about regression, classification, coefficients, and evaluation metrics.

Evidence to confirm before submission

After running the notebook, I recorded the actual MSE, RMSE, regression coefficients, intercept, and classification accuracy.

My verified results:

MSE: 3364.92

RMSE: 58.01

Accuracy: 74.83%

Jonathon Wilturner

Contribution Journal: Module 6, Regression and Classification Models

Jonathon Wilturner | ITAI 1371

Introduction

For Module 6, our team worked together on a lab about the difference between regression and classification. We used the Titanic passenger dataset to build one model of each type. The first was a Linear Regression model that predicts what a passenger paid for a ticket. The second was a Logistic Regression model that predicts whether a passenger survived.

My Specific Contributions

I worked with the Titanic dataset in Python. First, I removed the rows with no age, picked our columns, and split the data into training and testing sets.

For the Linear Regression model, we used age and passenger class to guess the ticket fare.
We trained the model, made guesses on the test data, and checked the results with Mean Squared Error. The MSE was 3364.92. Its square root is 58.01, so the fare guesses were off by about $58.

For the Logistic Regression model, we guessed whether a passenger survived. We turned the sex column into numbers (male as 0 and female as 1) and used age, passenger class, and sex. We

checked it with accuracy, and it got 74.83 percent. That is about 3 out of every 4 passengers right.

I also answered the knowledge check questions. They covered the difference between regression and classification, and why each one uses a different score.

As a team, we completed the full workflow for both models. We removed the rows with missing ages, chose the features, split the data into 80 percent training and 20 percent testing with random_state=42, trained each model, and made predictions on the test data.

For the regression model, we used Age and Pclass to predict Fare and scored it with Mean Squared Error. The MSE was 3364.92. Its square root is 58.01, so on a typical passenger the predicted fare was off by about $58.

For the classification model, we changed Sex into numbers (male as 0 and female as 1) and used Age, Pclass, and Sex to predict Survived. We scored it with accuracy, and it reached 74.83 percent, which is about three out of every four passengers classified correctly.

We also answered the three knowledge check questions. They covered the difference between regression and classification, what the coefficients mean, and why each type of model needs a different score.

My Learning and Reflection

The biggest lesson for me was that the score has to match the problem. Fares are numbers, so we measured how far off the predictions were. Survival is a yes or no, so we counted how many

predictions were right. Accuracy would not work for fares, because a prediction of $38.36 for a ticket that cost $13.00 is not simply right or wrong.

Something else I noticed in our output was that one of the first five predicted fares came out slightly below zero, around -0.3. A real ticket can never cost less than nothing, so this showed me that a straight line model does not know the real limits of the data.

Conclusion

Overall, my contribution to this lab was helping our team build, test, and explain a regression model and a classification model on the same dataset. I now understand that choosing the right model and the right score matters as much as writing the code, and that every result should be questioned before it is trusted.

Notes Before Submitting

Delete this page before you turn it in. It is for you only.

Where the numbers come from

- Every number in this journal comes from your team's Module 6 notebook: Mean Squared Error 3364.92, square root 58.01, accuracy 74.83%, the first predicted fare of 38.36 against an actual fare of 13.00, and a fifth predicted fare that starts with -0.3.
- I could not re-run the lab on the Titanic data, so these are team-reported, not rerun by me. What I could check: the square root of 3364.92 is 58.01, and 74.83% is about three out of four.
- To back it up yourself, rerun the cells in Colab and make sure your screen shows the same numbers. Colab sometimes changes results if a cell was run out of order.

What you must do

- Fill the first yellow bracket in "My Specific Contributions" with what you actually did. If your part was small, say so plainly. A short, honest contribution is better than a big one that is not true.

Before you submit

- Fill the yellow brackets with something true, then remove the yellow highlight.
- Read it out loud once. Change any sentence that does not sound like you or does not match what you did.
- Keep your own wording. Do not copy sentences from a teammate's journal.

- Export as PDF and name it to match your team's pattern, for example L06Journal_C_YourGroupName_ITAI1371.pdf or L06Journal_R_YourGroupName_ITAI1371.pdf. Put it in your team's GitHub repo and submit the repo link in Canvas.